EXPERIMENT 08 — LAB RECORD

# Configuring User Management and Role-Based Access Control (RBAC) in a Zoho Creator Application

Design and Implementation of User Permissions, Roles, Role Hierarchy, and Data Sharing for the Patient Appointment Scheduling System

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## Aim

To configure user management and role-based access control (RBAC) within the Patient Appointment Scheduling System built in Zoho Creator, by studying its existing permission sets, constructing a three-level role hierarchy, and configuring a custom data-sharing rule that grants the Medical Administrator role Read/Write access to Appointment Record data owned by the Receptionist role.

### Objectives

- To review the application's live dashboard and confirm the volume of records held across its four forms — Patient Department Links, Appointment Records, Department Masters, and Patient Profiles.
- To study the six permission sets available under User Permissions: Department Coordinator, Head of Department, Medical Administrator, Read, Receptionist, and Write.
- To construct the role hierarchy CEO → Medical Administrator → {Head of Department, Department Coordinator, Receptionist}.
- To configure a Data Sharing rule, Receptionist_to_Medical_Administrator, sharing Appointment Record data from the Receptionist role to the Medical Administrator role with Read/Write access.
- To understand how RBAC separates data ownership (who creates and holds a record) from data visibility (who else may view or edit it).

### Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- Patient Appointment Scheduling System application (created on Oct 05, 2026)
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

## Introduction / Theory

### User Management in Zoho Creator

User management decides who can access an application and what each person may see and do within it. Users are added to an application and associated with permission sets (profiles) and roles; together these control access to forms, reports, pages and records. In the Settings area, the User Permissions section groups these controls under three tabs: Permissions, Roles and Data Sharing.

### Role-Based Access Control (RBAC)

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RBAC is an access-control model in which rights are assigned to roles instead of individual users, and users acquire rights through the roles they hold. It simplifies administration, supports the principle of least privilege, improves data security, and keeps the access policy consistent and easy to audit.

### Key Concepts

| Concept | Description |
|---|---|
| **Users** | Individuals added to the application who sign in to work with its forms, reports and data. |
| **Permissions** | Named permission sets (profiles) defining what a user may do with components. This application lists Department Coordinator, Head of Department, Medical Administrator, Read, Receptionist, and Write. |
| **Roles** | Positions in the organisational structure (CEO, Medical Administrator, Head of Department, Department Coordinator, Receptionist) that determine a user's place in the hierarchy and the visibility of records. |
| **Role Hierarchy** | A tree in which each role reports to another. A higher role can generally view the records of roles beneath it; the reverse does not apply. |
| **Data Sharing** | Share rules that extend record visibility beyond the hierarchy, from one role or user to another, with a defined level of access (Read Only or Read/Write). |

### Application Specifications

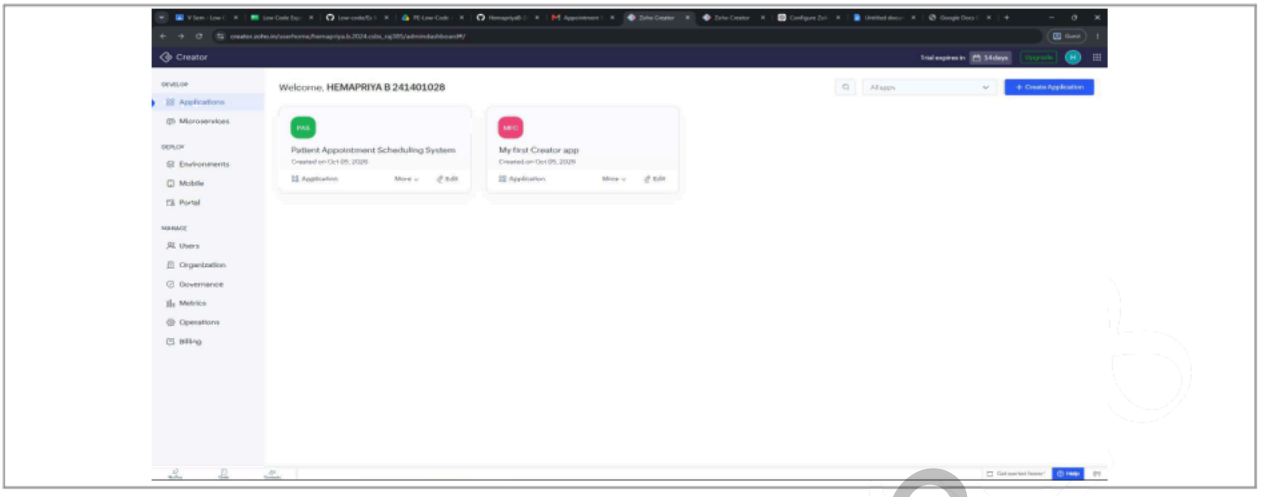
| Parameter | Detail |
|---|---|
| **Application Name** | Patient Appointment Scheduling System |
| **Primary Forms** | Patient Profile, Appointment Record, Department Master, Patient Department Link |
| **Dashboard** | Summary tiles (Total Patient Department Links, Total Appointment Records, Total Department Masters, Total Patient Profiles), Distribution by Priority Level chart, Review Status tiles, and Distribution by Created By chart |
| **Permission Sets** | Department Coordinator, Head of Department, Medical Administrator, Read, Receptionist, Write |
| **Role Hierarchy** | CEO → Medical Administrator → {Head of Department, Department Coordinator, Receptionist} |
| **Custom Sharing Rule** | Receptionist → Medical Administrator, on Appointment Record, Read/Write access |

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## Procedure

**1. Sign in and locate the application** — Signing in to Zoho Creator opens the home page listing the available applications; the Patient Appointment Scheduling System card (created Oct 05, 2026) is located among them.

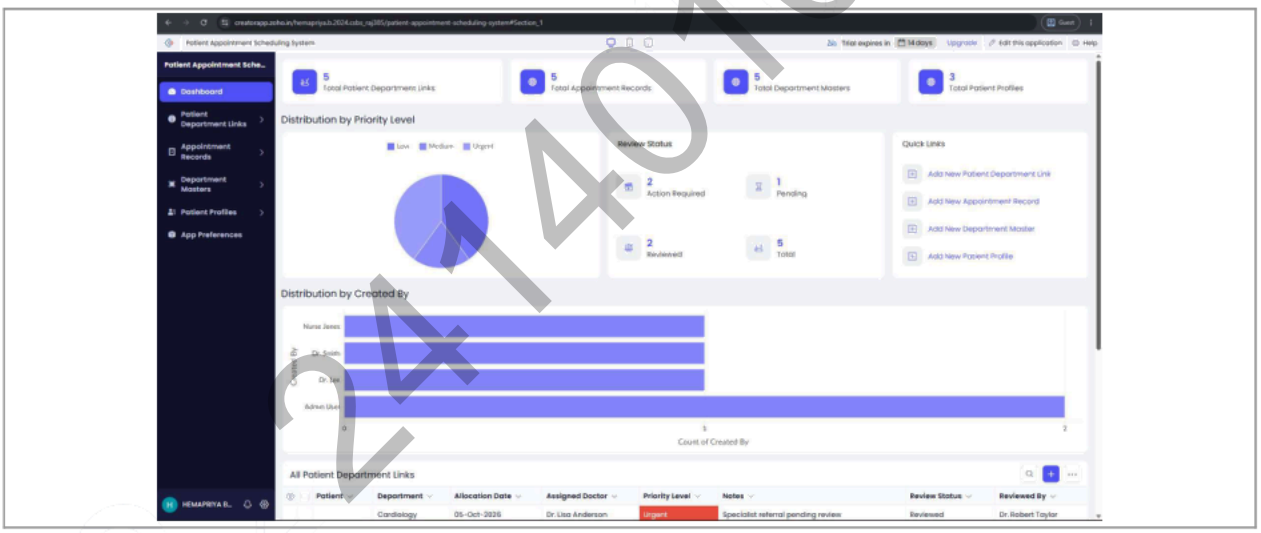
***Figure 8.1: Zoho Creator Home — Patient Appointment Scheduling System***




The home page lists both applications in the account, with Patient Appointment Scheduling System shown as the first card, alongside Application and Edit shortcuts.

**2. Open the dashboard** — Opening the application loads its Dashboard, showing the All Patient Department Links report alongside summary tiles and participant/priority charts.

***Figure 8.2: Patient Appointment Scheduling System — Dashboard (Access Mode)***

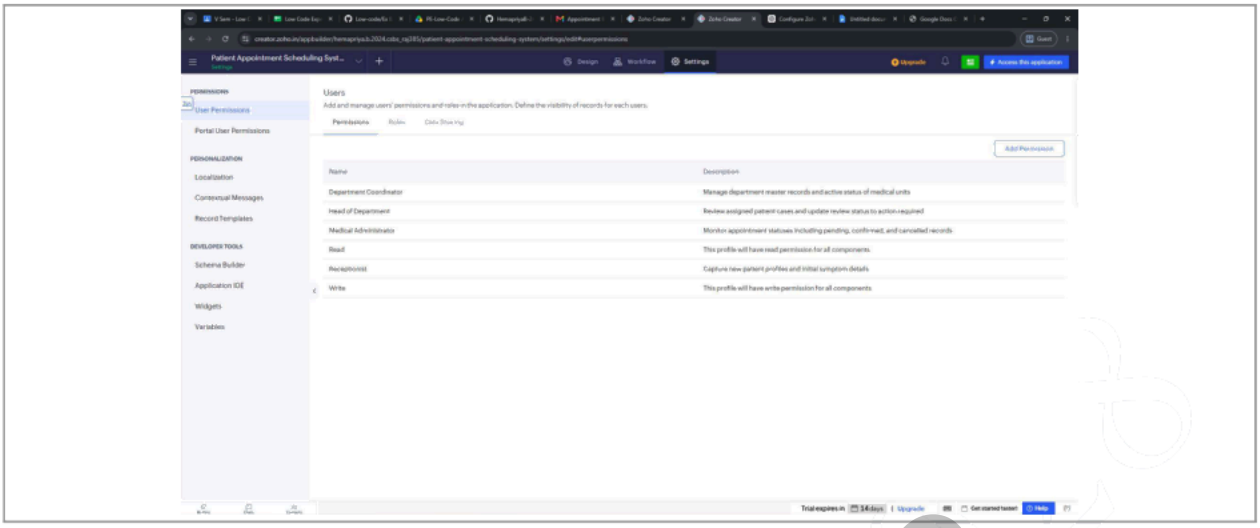



The dashboard shows four summary tiles (5 Patient Department Links, 5 Appointment Records, 5 Department Masters, 3 Patient Profiles), a Distribution by Priority Level pie chart, Review Status counts, Quick Links, and a Distribution by Created By bar chart listing Nurse Jones, Dr. Smith, Dr. Lee and Admin User.

**3. Enter Edit mode and open Settings** — Selecting Edit this application switches to Edit mode, and the Settings tab is selected from the top navigation bar (Design, Workflow, Settings).

**4. Open User Permissions** — Under the PERMISSIONS group, User Permissions is selected, opening the Users page with the Permissions, Roles and Data Sharing tabs.

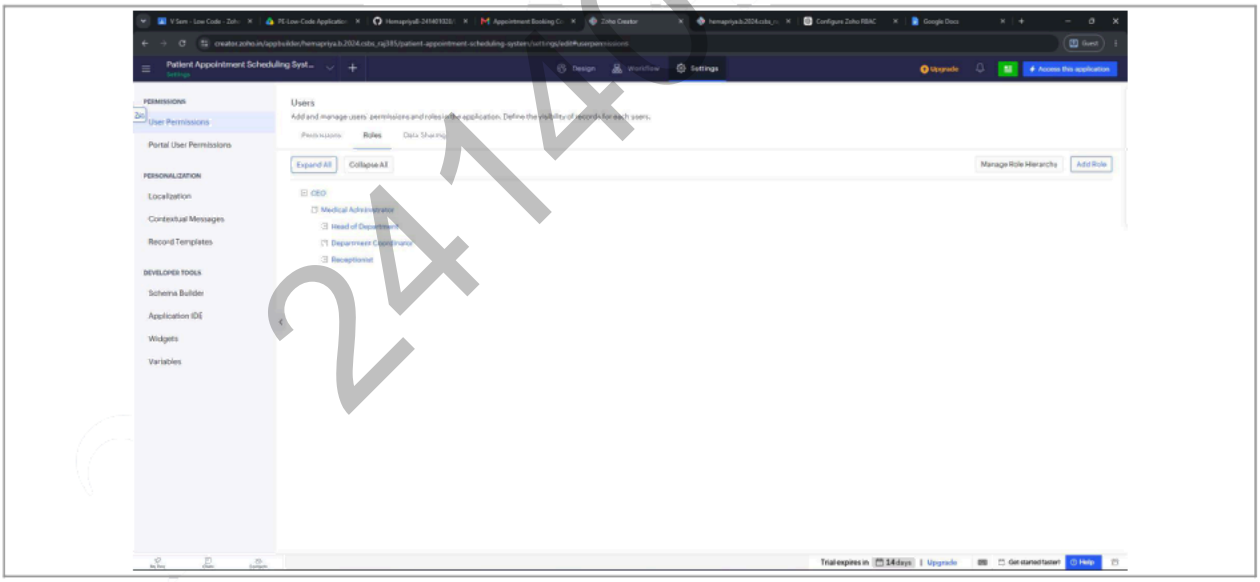
***Figure 8.3: User Permissions — Permissions Section***




The Permissions tab lists six permission sets: Department Coordinator (manages department master records and medical unit status), Head of Department (reviews assigned cases), Medical Administrator (monitors appointment statuses), Read, Receptionist (captures new patient profiles and symptoms), and Write — each with a short description and an Add Permission option.

**5. Build the role hierarchy** — On the Roles tab, the role tree is arranged as CEO → Medical Administrator → Head of Department, Department Coordinator, and Receptionist, using Expand All, Collapse All, Manage Role Hierarchy and Add Role as needed.

***Figure 8.4: Roles and Role Hierarchy***

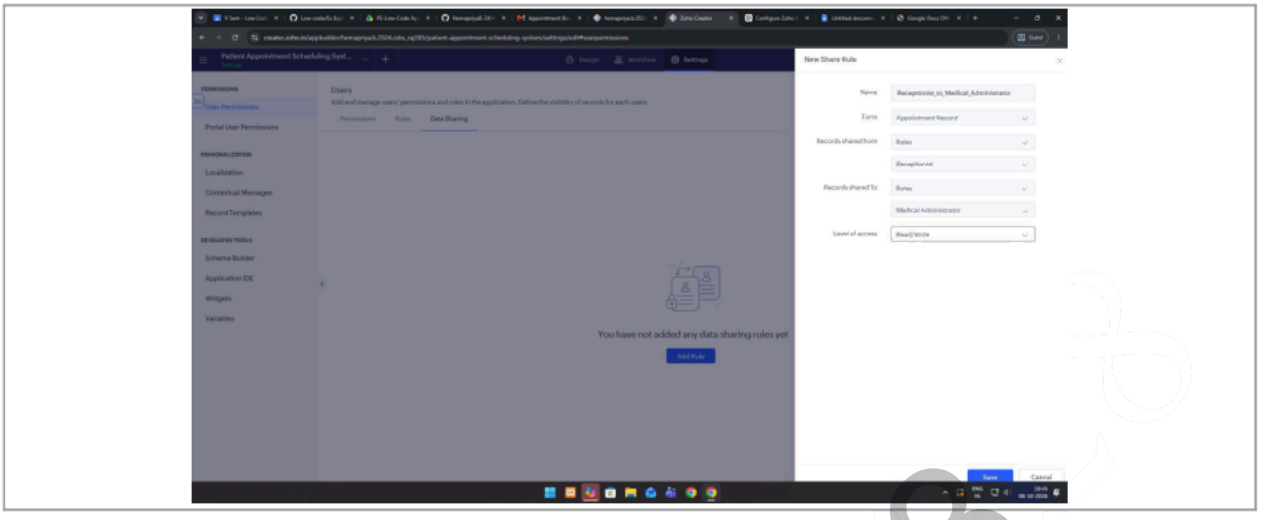



The Roles tab shows the completed hierarchy: CEO at the top, Medical Administrator reporting to it, and Head of Department, Department Coordinator, and Receptionist all reporting to Medical Administrator as sibling roles.

**6. Create a new share rule** — On the Data Sharing tab, New Share Rule is selected, the rule is named Receptionist_to_Medical_Administrator, the Form is set to Appointment Record, and Records shared from is set to Roles → Receptionist.

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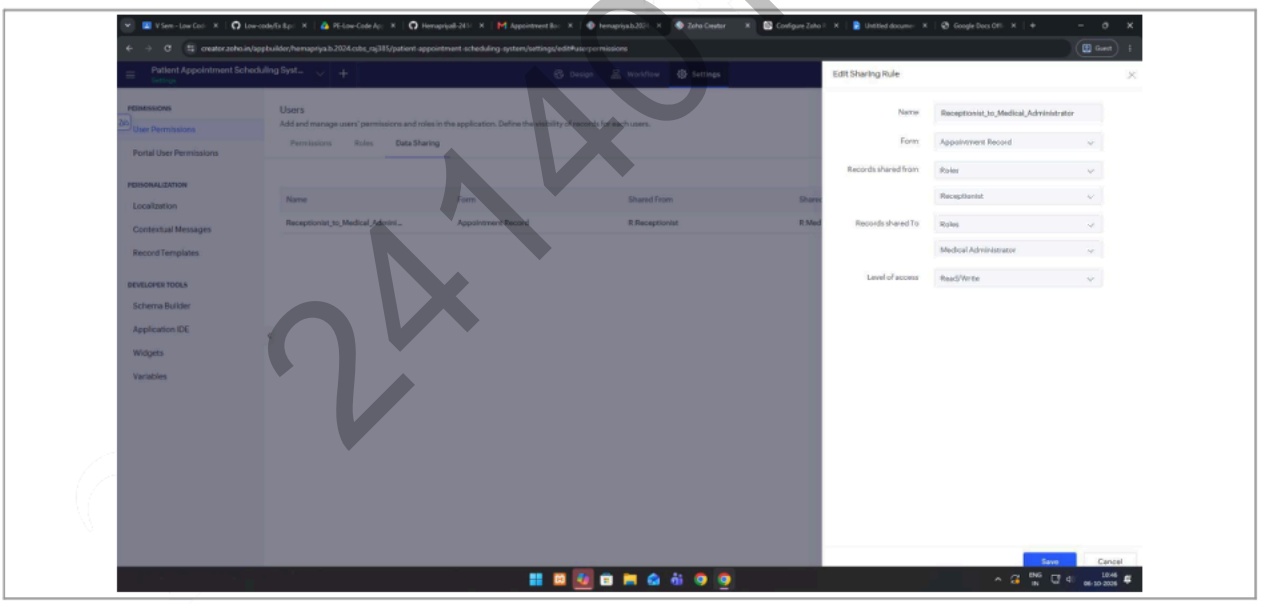
***Figure 8.5: New Share Rule — Rule Configuration***




The New Share Rule panel shows the rule Receptionist_to_Medical_Administrator on the Appointment Record form, with Records shared from set to Roles → Receptionist, Records shared To set to Roles → Medical Administrator, and Level of access set to Read/Write.

**7. Save and verify the rule** — Selecting Save adds the rule to the Data Sharing list; reopening it with Edit Sharing Rule confirms every field was stored exactly as configured.

***Figure 8.6: Configured Data Sharing Rule (Receptionist → Medical Administrator, Read/Write)***




The Data Sharing list now shows the saved rule — Form: Appointment Record, Shared From: R:Receptionist, Shared To: R:Medical Administrator — and the Edit Sharing Rule panel confirms the Read/Write access level.

## Data Sharing

The application follows a layered model: Form (Patient Profile / Appointment Record / Department Master / Patient Department Link) → Dashboard → Security Layer (Permissions, Roles, Data Sharing). The role hierarchy runs from the CEO (overall oversight) down through Medical Administrator

(department-wide monitoring) to Head of Department, Department Coordinator, and Receptionist (day-to-day data entry and case handling). By default the hierarchy gives only upward visibility: higher roles see the records of the roles below them, but not the reverse.

The custom data-sharing rule shares Appointment Record data from the Receptionist role to the Medical Administrator role at Read/Write level. Because the Receptionist is the role that actually captures new patient profiles and initial appointment details, this rule gives the Medical Administrator — who is positioned above Receptionist in the hierarchy and would already see these records by inheritance — an explicit, auditable Read/Write channel to update appointment status (pending, confirmed, cancelled) directly, rather than relying solely on implicit hierarchy visibility. It keeps the scope directional (Receptionist → Medical Administrator), documents exactly who can modify what in the Data Sharing list, and lets the Medical Administrator act on records without needing elevated, system-wide Write permission.

### RBAC Working

| Level | Role | Intended Function |
|---|---|---|
| **1 (Top)** | **CEO** | Overall authority; able to view the records of every role reporting to it. |
| **2** | **Medical Administrator** | Monitors appointment statuses across departments; gains Read/Write access to Receptionist-owned Appointment Records through the custom share rule. |
| **3** | **Head of Department / Department Coordinator / Receptionist** | Handle day-to-day data entry and case review — reviewing assigned cases, managing department masters, and capturing new patient profiles respectively. |

- Users in the Receptionist, Department Coordinator, and Head of Department roles work with records according to their assigned permission set.
- Because these three roles all report to Medical Administrator, who reports to CEO, higher roles can view the records of the roles below them by default.
- Lower roles do not automatically gain access to higher-role records; the Read/Write share rule grants the Medical Administrator a controlled, explicit channel to Receptionist-owned Appointment Records.
- The permission sets (Department Coordinator, Head of Department, Medical Administrator, Read, Receptionist, Write) further limit the specific operations each user may carry out on each component.

## Result

- The Patient Appointment Scheduling System application, with its Patient Profile, Appointment Record, Department Master and Patient Department Link forms, was opened and its dashboard reviewed.
- The six permission sets — Department Coordinator, Head of Department, Medical Administrator, Read, Receptionist, and Write — were studied under User Permissions.

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- The role hierarchy CEO → Medical Administrator → {Head of Department, Department Coordinator, Receptionist} was created.
- A data-sharing rule from Receptionist to Medical Administrator with Read/Write access on Appointment Record was configured and saved.

## Conclusion

The experiment demonstrated how permissions, roles, role hierarchy and data sharing together form a role-based access control mechanism in Zoho Creator. Structured forms with typed fields and a role-aware dashboard replace unstructured, hard-to-audit record keeping with clean, consistent, and analysable data. The four-level hierarchy and the directional Read/Write share rule keep information channels controlled: the Medical Administrator gains exactly the access needed to monitor and update appointment statuses, while the CEO retains overall oversight and the Receptionist, Department Coordinator and Head of Department continue to own their respective records. The result is a secure, auditable and scalable access-control framework for the patient appointment application.

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